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Vehicle console assembly

Abstract

A vehicle console assembly includes a body that defines a storage space. The body includes an upper rim. A support assembly is operable between a first position at one end of the upper rim and a second position at a second end of the upper rim. The support assembly is operable between a folded armrest condition and a deployed tray condition. An actuation assembly is operably coupled to the support assembly. The actuation assembly includes at least one cable having coils. The at least one cable is coupled to the support assembly. A motor is configured to engage the coils of the at least one cable to translate the support assembly between the first position and the second position.

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Background/Summary

FIELD OF THE DISCLOSURE

(1) The present disclosure generally relates to a console assembly. More specifically, the present disclosure relates to a console assembly for a vehicle.

BACKGROUND OF THE DISCLOSURE

(2) Many vehicles include center consoles. The center consoles often provide a storage cavity for housing and storing items. Additionally, the center consoles often have a cover that includes a padded layer.

SUMMARY OF THE DISCLOSURE

(3) According to at least one aspect of the present disclosure, a vehicle console assembly includes a body that defines a storage space. The body includes an upper rim. A support assembly is operable between a first position at a first end of the upper rim and a second position at a second end of the upper rim. The support assembly is operable between a folded armrest condition and a deployed tray condition. An actuation assembly is operably coupled to the support assembly. The actuation assembly includes at least one cable having coils. The at least one cable is coupled to the support assembly. A motor is configured to engage the coils of the at least one cable to translate the support assembly between the first position and the second position.

(4) According to another aspect of the present disclosure, a power tray system for a center console includes a track. A flocked drive cable extends through the track. A motor has a drive gear configured to engage coils of the flocked drive cable. A support assembly is coupled to the flocked drive cable. The drive gear is configured to engage the coils of the flocked drive cable to drive movement of the flocked drive cable relative to the track and, consequently, translate the support assembly between first and second positions relative to the track. The support assembly is operable between a folded armrest condition and deployed tray condition.

(5) According to another aspect of the present disclosure, a center console includes a body. A track is coupled to the body. A support assembly has side panels rotatably coupled to a connector panel. The side panels are operable between a folded condition over the connector panel and a deployed condition forming a continuous surface with the connector panel. An actuation assembly is operably coupled to the support assembly and the track. The actuation assembly includes a cable extending through the track and coupled to the support assembly. A motor has a drive gear. The drive gear is configured to engage the cable to translate the support assembly between first and second positions relative to the body.

(6) These and other aspects, objects, and features of the present disclosure will be understood and appreciated by those skilled in the art upon studying the following specification, claims, and appended drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The following is a description of the figures in the accompanying drawings. The figures are not necessarily to scale, and certain features and certain views of the figures may be shown exaggerated in scale or in schematic in the interest of clarity and conciseness.

(2) In the drawings:

(3) FIG. 1 is a side perspective view of a console assembly within a vehicle, with a support assembly in a first position and in a folded condition, according to the present disclosure;

(4) FIG. 2 is a side perspective view of a console assembly within a vehicle, with a support assembly in a first position and in a folded condition, according to the present disclosure;

(5) FIG. 3 is a side perspective view of a console assembly within a vehicle, with a support assembly in an intermediate position and in a folded condition, according to the present disclosure;

(6) FIG. 4 is a side perspective view of a console assembly within a vehicle, with a support

- assembly in a second position and in a folded condition, according to the present disclosure;
- (7) FIG. 5 is a side perspective view of a support assembly for a console assembly operably coupled with an actuation assembly, with the support assembly in a first position and in a deployed condition, according to the present disclosure;
- (8) FIG. 6 is a partial cross-sectional view of a support assembly engaging an actuation assembly within a track for a console assembly, according to the present disclosure;
- (9) FIG. 7 is a partial side perspective view of an actuation assembly for a power tray system of a console support assembly, according to the present disclosure;
- (10) FIG. 8 is a partial cross-sectional view of a motor engaging cables for a power tray system of a console assembly, according to the present disclosure
- (11) FIG. 9 is a block diagram of a power tray system, according to the present disclosure; and
- (12) FIG. 10 is a side perspective view of a console assembly within a vehicle, with a support assembly in a second position and in a deployed condition, according to the present disclosure.

DETAILED DESCRIPTION

(13) Additional features and advantages of the presently disclosed device will be set forth in the detailed description which follows and will be apparent to those skilled in the art from the description, or recognized by practicing the device as described in the following description, together with the claims and appended drawings.

(14) For purposes of description herein, the terms “upper,” “lower,” “right,” “left,” “rear,” “front,” “vertical,” “horizontal,” and derivatives thereof shall relate to the concepts as oriented in FIG. 1. However, it is to be understood that the concepts may assume various alternative orientations, except where expressly specified to the contrary. It is also to be understood that the specific devices and processes illustrated in the attached drawings, and described in the following specification are simply exemplary embodiments of the inventive concepts defined in the appended claims. Hence, specific dimensions and other physical characteristics relating to the embodiments disclosed herein are not to be considered as limiting, unless the claims expressly state otherwise.

(15) As used herein, the term “and/or,” when used in a list of two or more items, means that any one of the listed items can be employed by itself, or any combination of two or more of the listed items, can be employed. For example, if a composition is described as containing components A, B, and/or C, the composition can contain A alone; B alone; C alone; A and B in combination; A and C in combination; B and C in combination; or A, B, and C in combination.

(16) As used herein, the term “about” means that amounts, sizes, formulations, parameters, and other quantities and characteristics are not and need not be exact, but may be approximate and/or larger or smaller, as desired, reflecting tolerances, conversion factors, rounding off, measurement error and the like, and other factors known to those of skill in the art. When the term “about” is used in describing a value or an end-point of a range, the disclosure should be understood to include the specific value or end-point referred to. Whether or not a numerical value or end-point of a range in the specification recites “about,” the numerical value or end-point of a range is intended to include two embodiments: one modified by “about,” and one not modified by “about.” It will be further understood that the end-points of each of the ranges are significant both in relation to the other end-point, and independently of the other end-point.

(17) As used herein the terms “the,” “a,” or “an,” mean “at least one,” and should not be limited to “only one” unless explicitly indicated to the contrary. Thus, for example, reference to “a component” includes embodiments having two or more such components unless the context clearly indicates otherwise.

(18) In this document, relational terms, such as first and second, top and bottom, and the like, are used solely to distinguish one entity or action from another entity or action, without necessarily requiring or implying any actual such relationship or order between such entities or actions. The terms “comprises,” “comprising,” or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises a list of

elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. An element preceded by “comprises . . . a” does not, without more constraints, preclude the existence of additional identical elements in the process, method, article, or apparatus that comprises the element.

(19) With reference to FIGS. 1-10, reference numeral **10** generally designates a console assembly for a vehicle **12** that includes a body **14** defining a storage space **16**. The body **14** includes an upper engagement portion or rim **18**, which defines an opening **20** into the storage space **16**. A support assembly **22** is operable between a first position **24** at a first end **26** of the upper rim **18** and a second position **28** at a second end **30** of the upper rim **18**. The support assembly **22** is also operable between a folded armrest condition **32** and a deployed tray condition **34**. An actuation assembly **36** is operably coupled to the support assembly **22**. The actuation assembly **36** includes at least one cable **38** having coils **40**. The at least one cable **38** is coupled to the support assembly **22**. The actuation assembly **36** also includes a motor **42** configured to engage the coils **40** of the at least one cable **38** to translate the support assembly **22** between the first and second positions **24**, **28**.

(20) Referring to FIGS. 1 and 2, the vehicle **12** includes an interior compartment **50** with at least one seating row **52**. Any practical number of seating rows **52** may be included in the vehicle **12** without departing the teachings herein. The illustrated seating row **52** is a front seating row **52** that includes a driver seat **54** and a passenger seat **56** that are separated by the console assembly **10** (e.g., the center console **10**).

(21) The console assembly **10** includes the body **14** having the first end **26**, which is generally a vehicle-rearward end, and the second end **30**, which is generally a vehicle-forward end. The vehicle-rearward first end **26** includes a support surface **66**, which is generally flat or smooth. The surface may be substantially rigid to provide support for items or may include a padded or cushioned material. At the vehicle-forward second end **30**, the console assembly **10** defines cup holders **68**.

(22) The body **14** defines the storage space **16** and a receiving feature **70** extending at least partially into the storage space **16**. The receiving feature **70** is generally disposed proximate to the cup holders **68** and may be configured for receiving items such as navigations, electronics, etc. The receiving feature **70** may extend from within the storage space **16** to a top surface of the upper rim **18**, as illustrated in FIG. 1, or may extend beyond the top surface of the upper rim **18**, as illustrated in FIG. 2.

(23) As illustrated in FIG. 1, the storage space **16** may be closed by an inner door **72** to prevent access to the storage space **16**. The inner door **72** may be configured to rotate or otherwise open to allow access to the storage space **16** and close to prevent access to the storage space **16**. In such examples, the inner door **72** may divide the storage space **16**, enclosing some or most of the storage space **16** below the inner door **72**. The area of the storage space **16** above the inner door **72** may be an open cavity, which can be used for storing items.

(24) In examples with the inner door **72**, the inner door **72** is disposed below the support assembly **22**. Accordingly, the support assembly **22** is configured to translate over the inner door **72**. It is contemplated that the support assembly **22** does not substantially impinge on the movement of the inner door **72** for accessing the storage space **16**.

(25) As illustrated in FIG. 2, the storage space **16** may be a substantially open cavity below the support assembly **22**, without the inner door **72** (FIG. 1). In such examples, the support assembly **22** may partially cover the storage space **16**. The storage space **16** may be accessible around the support assembly **22**. Each of the enclosed storage space **16** with the inner door **72** and the open storage space **16** may be utilized with each configuration of the receiving feature **70** or without the receiving feature **70**.

(26) Referring still to FIG. 2, as well as FIGS. 3 and 4, the console assembly **10** includes the support assembly **22**, which is configured to translate between the first position **24** at the first end **26**, illustrated in FIG. 2, and the second position **28** at the second end **30**, illustrated in FIG. 4. The

support assembly **22** is configured to translate over the storage space **16** and/or the inner door **72**, depending on the configuration of the console assembly **10**. At least a portion of the storage space **16** is accessible when the support assembly **22** is in the first position **24**, in the second position **28**, or therebetween. Further, the support assembly **22** may be configured to be stopped in an intermediate position between the first and second positions **24**, **28**, such as the position illustrated in FIG. 3.

(27) Referring still to FIGS. 2-4, as well as FIGS. 5-7, the support assembly **22** is driven between the first and second positions **24**, **28** by the actuation assembly **36**. The actuation assembly **36** extends along an inner surface of and/or within the upper rim **18**. Generally, the actuation assembly **36** extends along the upper rim **18** about at least a portion of the opening **20** to the storage space **16**.

(28) The actuation assembly **36** engages a track **80** on the body **14** and the support assembly **22** to drive movement of the support assembly **22** along the track **80**. The track **80** includes opposing track portions **82**, **84** that extend in the fore-aft direction on the upper rim **18**. As illustrated in FIG. 5, the track **80** may include an elevation change, such that a rear portion of the track **80** is lower than a front portion. This may be advantageous for positioning the support assembly **22** for the folded armrest condition **32** relative to the deployed tray condition **34**. The track **80** may have horizontal portions at each end and a sloped portion therebetween, as illustrated in FIG. 5, or have a consistent slope. It is also contemplated that the track **80** may be linear without the elevation change or slope in an opposing direction (i.e., front portion lower than rear portion) depending on the configuration of the console assembly **10**.

(29) The support assembly **22** includes a connector panel **90** extending between the two track portions **82**, **84** on opposing sides **92**, **94** of the upper rim **18** and over the storage space **16**. Engagement portions **96** extend from a bottom surface **100** of the connector panel **90**, proximate to each side **92**, **94**, and into the storage space **16** to engage the track **80**. The engagement portions **96** each include a drive feature **102**, which are configured to engage the track **80** and the actuation assembly **36**.

(30) Each track portion **82**, **84** is configured to guide movement of the respective drive feature **102** of the support assembly **22**, as well as the drive cable **38** and a return cable **104** of the actuation assembly **36**. Accordingly, each track portion **82**, **84** includes a guide rail **106**, which slidably receives the drive features **102** of the engagement portions **96**, and two channels **108**, **110** for the cables **38**, **104**. The first channel **108** for the drive cable **38** is open to the guide rail **106**, allowing for a connection between the drive features **102** and the drive cable **38**. The drive cable **38** is overmolded to each drive feature **102**, forming a fixed connection therebetween. This fixed connection allows the support assembly **22** to move with the drive cable **38**.

(31) The return cable **104** extends through the second channel **110** and is separated from the guide rail **106**, allowing the return cable **104** to move relative to the support assembly **22**. The return cable **104** is generally excess cable length caused by movement of the drive cable **38** based on the configuration of the drive cable **38** and the position of the support assembly **22**. The return cable **104** may be integrally formed with the drive cable **38** or separate cables **38**, **104**. While generally not directly fixed to the support assembly **22** within the track **80**, the return cable **104** assists in moving the drive cable **38** and, consequently, the support assembly **22**.

(32) Referring still to FIGS. 2-8, the actuation assembly **36** extends around a portion of a perimeter of the opening **20** into the storage space **16**. In the illustrated example in FIG. 5, the actuation assembly **36** forms a U-shape that extends along both sides **92**, **94** and the second end **30** of the upper rim **18**. It is also contemplated that the actuation assembly **36** may extend along the first end **26** of the upper rim **18** or the entire perimeter without departing from the teachings herein. As illustrated in FIG. 5, the motor **42** is disposed in or at the vehicle-forward second end **30** of the console assembly **10** and is configured to engage each of the drive cable **38** and the return cable **104**.

(33) As illustrated in FIGS. 7 and 8, the motor **42** includes a drive gear **118**, which engages the

drive and return cables **38, 104**. The cables **38, 104** are generally configured as flocked cables **38, 104** having the coils **40**. The coils **40** are arranged along at least a substantial portion of the length or the entire length of each of the cables **38, 104**. The drive gear **118** engages the coils **40** to form a gear train, operating to drive the movement of the cables **38, 104** and, consequently, the support assembly **22**. Generally, rotation of the drive gear **118** in a clockwise direction drives the support assembly **22** in the vehicle-forward direction from the first position **24** to the second position **28**. Rotation of the drive gear **118**, a counterclockwise direction drives the support assembly **22** in the vehicle-rearward direction from the second position **28** to the first position **24**.

(34) Referring still to FIGS. **7** and **8**, as well as to FIG. **9**, the motor **42** is operably coupled with an electronic control unit **120** (ECU), which may be the control unit for the vehicle **12** or may be a designated control unit for the console assembly **10**. The electronic control unit **120** has a processor **122**, a memory **124**, and other control circuitry. Instructions or routines **126** are stored in the memory **124** and executable by the processor **122**.

(35) The ECU **120** is also in communication with a Hall sensor **128**, which is communicatively coupled with the motor **42**. The Hall sensor **128** is configured to sense or count the number of rotations of the motor **42**, allowing the ECU **120** to determine the position of the support assembly **22**. The ECU **120** is configured to monitor the sensed information from the Hall sensor **128** to stop the support assembly **22** in the first or second position **24, 28**, as well as determine if there is an obstruction in the movement path. If there is an object in the movement path, the Hall sensor **128** is configured to sense and communicate a change in value and the ECU **120** may operate to stop or reverse the rotation of the motor **42**. Accordingly, after detection of an obstruction, the support assembly **22** may be stopped or moved in an opposing direction away from the obstruction.

(36) In various examples, the ECU **120** is configured to stop the support assembly **22** in an intermediate position between the first and second positions **24, 28**. These intermediate positions may be stored within the ECU **120** and may be determined based on rotations sensed by the Hall sensor **128**.

(37) The ECU **120** may also be communicatively coupled to a user interface **130**, which may be a screen, button, or other interfaces within the vehicle **12** or part of a remote device. A user may control the position of the support assembly **22** by inputting a position for the support assembly **22** through the user interface **130**. The ECU **120** is configured to receive the input and adjust the support assembly **22** accordingly.

(38) Referring again to FIGS. **2** and **4**, as well as FIG. **10**, the support assembly **22** is also operable between the folded armrest condition **32** and the deployed tray condition **34**. Generally, the support assembly **22** is disposed in the folded armrest condition **32** when in the first position **24** and during movement and in the deployed tray condition **34** when in the second position **28**. However, the support assembly **22** may be configured to be in the folded armrest condition **32** or the deployed tray condition at any position along the upper rim **18** without departing from the teachings herein.

(39) The support assembly **22** includes the connector panel **90**, which spans over the storage space **16** between the first and second sides **92, 94** of the console assembly **10**. Accordingly, the connector panel **90** extends the width of the storage space **16**. Two side panels **140, 142** are rotatably coupled to the connector panel **90** and are configured to rotate between the folded condition **32** and the deployed condition **34**. The side panels **140, 142** are coupled to opposing lateral sides of the connector panel **90** proximate the sides **92, 94** of the body **14**, respectively.

(40) When the first and second side panels **140, 142** are disposed in the folded condition **32**, the first and second side panels **140, 142** are disposed over and rest on top of the connector panel **90** adjacent to one another. In the folded condition **32**, the width of the support assembly **22** is substantially equal to the width of the connector panel **90**. The side panels **140, 142** may or may not cover the entirety of the top surface of the connector panel **90** when in the folded condition **32**.

(41) The side panels **140, 142** also include a cushion **144**, which may be a pad, padded layer, etc. When the side panels **140, 142** are in the folded condition **32**, the cushions **144** are on upper

surfaces of the side panels **140**, **142** and accessible for the user to rest his or her arm. Accordingly, the side panels **140**, **142** are configured to operate as armrests for passengers in the driver and passenger seats **54**, **56** of the seating row **52**.

(42) When the side panels **140**, **142** are opened into the deployed tray condition **34**, the side panels **140**, **142** and the connector panel **90** form a substantially continuous and generally flat surface configured to operate as a tray or table. Accordingly, the width of the support assembly **22** is the combined width of the connector panel **90** and both side panels **140**, **142** adjacent to one another. The cushions **144** are also disposed on a lower surface of the side panels **140**, **142** when the side panels **140**, **142** are deployed, providing for a more rigid and more stable surface for the tray or table.

(43) In the deployed condition **34**, the support assembly **22** may have a curved vehicle-rearward edge, which may provide more space for the passengers in the seating row **52** when the side panels **140**, **142** are deployed. While both side panels **140**, **142** are illustrated in the deployed condition **34** in FIG. **10**, it is contemplated that one side panel **140** may be disposed in the folded condition **32** and the other side panel **142** may be in the deployed condition **34** without departing the teachings herein.

(44) The tray or table formed by the support assembly **22** may have a width in a range between 400 mm and 600 mm in the cross-car direction. Generally, the tray may be a sufficient size to support electronics, such as a tablet or laptop, to provide a mobile workstation or hold a meal. The support assembly **22** may be any practicable size for providing the armrest and the tray.

(45) Referring again to FIGS. **6** and **10**, the side panels **140**, **142** are rotatably coupled to the connector panel **90**. In various examples, the side panels **140**, **142** may be coupled to the connector panel **90** via hinge assemblies **146**, respectively. In such examples, the hinge assemblies **146** may form part of the continuous tray surface. In such examples, the hinge assemblies **146** may be coplanar with the connector panel **90** and the side panels **140**, **142** in the deployed condition **34**, may be between or within the connector panel **90** and the side panels **140**, **142**, or may be below the tray surface. Additionally or alternatively, the side panels **140**, **142** may be coupled to the connector panel **90** via living hinges. Accordingly, the engagement between the side panels **140**, **142** and the connector panel **90** may not substantially impinge the tray surface of the support assembly **22**.

(46) In various aspects, the side panels **140**, **142** may be manually rotated between the folded and deployed conditions **32**, **34**. The user may rotate the side panels **140**, **142** when the support assembly **22** is at any position. Additionally or alternatively, an actuator may be configured to rotate the side panels **140**, **142** between the folded and deployed conditions **32**, **34**. Further, the side panels **140**, **142** may be locked in certain positions or during movement of the support assembly **22** and unlocked at other positions. In such examples, the ECU **120** may be configured to determine the position of the support assembly **22** based on the Hall sensor **128** and unlock the support assembly **22** to allow the support assembly **22** to be deployed.

(47) Referring to FIGS. **1-10**, the support assembly **22** is configured to provide the armrest and the tray or table for the passengers in the seating row **52**. When the support assembly **22** is operating as the tray in the deployed condition **34**, the support assembly **22** may provide a mobile workstation, which may be advantageous for providing space for an occupant of the vehicle **12** to work while the vehicle **12** is parked and stationary, or provide a place to eat lunch while at a worksite. In the current regulatory framework, it is contemplated that the tray configuration of the support assembly **22** for providing a work station or lunch table is used when the vehicle **10** is parked and stationary. However, it is anticipated that technology and the regulatory framework may evolve in the future to where using the support assembly **22** operated as a tray in a moving vehicle becomes safe and permissible. Accordingly, the powered tray system of the console assembly **10** may be utilized in accordance with current safety and regulatory framework and guidelines.

(48) The support assembly **22** is integrated into the console assembly **10** and provides a convenient

and mobile workspace. The support assembly **22** is adjusted between the first and second positions **24**, **28** by the actuation assembly **36** and, in at least some examples, between the folded and deployed conditions **32**, **34** by the actuator. Accordingly, the console assembly **10** includes the support assembly **22** providing a power tray or table system. In the illustrated example, the console assembly **10** is disposed between the driver seat **54** and the passenger seat **56** of the front seating row **52**. However, it is contemplated that the console assembly **10** may be positioned in any practicable location within the interior compartment **50** of the vehicle **12** without departing from the teachings herein.

(49) The vehicle **12** is a wheeled motor vehicle **12**, which may be a sedan, a sport-utility vehicle, a truck, a van, a crossover, and/or other styles of vehicle **12**. The vehicle **12** may be a manually operated vehicle **12** (e.g., operated with a human driver), a fully autonomous vehicle **12** (e.g., operated without a human driver), or a partially autonomous vehicle **12** (e.g., operated with or without a human driver). Additionally, the vehicle **12** would be utilized for personal and/or commercial purposes, such as ride providing services (e.g., chauffeuring) and/or ridesharing services.

(50) Use of the present device may provide a variety of advantages. For example, the console assembly **10** includes an integrated power armrest-tray system. Additionally, the support assembly **22** is driven by the actuation assembly **36** between the first position **24** the second position **28** relative to the body **14** of the console assembly **10**. Also, the support assembly **22** is configured to translate over the storage space **16**, which may provide different access points for the storage space **16**. Moreover, the actuation assembly **36** may use flocked cables **38**, **104** and a drive gear **118** to form a gear train to translate the support assembly **22** fore and aft. Further, the support assembly **22** is operable between the folded condition **32** to provide an armrest for passengers and the deployed condition **34** to provide a table or tray surface. Also, the side panels **140**, **142** may be manually adjusted or actuated between the folded and the deployed conditions **32**, **34**. Moreover, the support assembly **22** and the actuation assembly **36** are integrated into the console assembly **10**. Additional benefits or advantages may be realized and/or achieved.

(51) According to various examples, a vehicle console assembly includes a body that defines a storage space. The body includes an upper rim. A support assembly is operable between a first position at one end of the upper rim and a second position at a second end of the upper rim. The support assembly is operable between a folded armrest condition and a deployed tray condition. An actuation assembly is operably coupled to the support assembly. The actuation assembly includes at least one cable having coils. The at least one cable is coupled to the support assembly. A motor is configured to engage the coils of the at least one cable to translate the support assembly between the first position and the second position. Embodiments of the present disclosure may include one or a combination of the following features: the support assembly includes a connector panel extending between opposing sides of the rim; a first side panel rotatably coupled to a first side of the connector panel; a second side panel rotatably coupled to a second side of the connector panel; the first and second side panels are disposed on the connector panel in the folded armrest condition and disposed adjacent to the connector panel in the deployed tray condition; the support assembly is in the folded armrest condition when in the first position on the upper rim and in the deployed tray condition when in the second position on the upper rim; the at least one cable includes a drive cable and a return cable; the support assembly includes an engagement portion having a drive feature, and wherein the drive feature is fixedly coupled to the drive cable; a track extending along the upper rim of the body; at least one cable extends through a channel defined by the track; the support assembly includes an engagement portion extending into the storage space; the engagement portion includes a drive feature slidably received in a guide rail of the track; an electronic control unit communicatively coupled to the actuation assembly; the actuation assembly includes a Hall sensor configured to sense rotations of the motor; and the electronic control unit is configured to monitor the sensed rotations to determine a position of the support relative to the upper rim

(52) According to various examples, a power tray system for a center console includes a track. A flocked drive cable extends through the track. A motor has a drive gear configured to engage coils of the flocked drive cable. A support assembly is coupled to the flocked drive cable. The drive gear is configured to engage the coils of the flocked drive cable to drive movement of the flocked drive cable relative to the track and, consequently, translate the support assembly between first and second positions relative to the track, and wherein the support assembly is operable between a folded armrest condition and deployed tray condition. Embodiments of the present disclosure may include one or a combination of the following features: the support assembly includes side panels; each side panel includes a cushion on one side thereof; the cushions are on upper surfaces of the side panels when the side panels are in the folded armrest condition; the cushions are on lower surfaces of the side panels when the side panels are in the deployed tray condition; the support assembly includes: a connector panel and side panels rotatably coupled to the connector panel; the side panels are disposed on the connector panel in the folded armrest condition and adjacent to the connector panel in the deployed tray condition; the track includes a first track portion and a second track portion; the connector panel extends between the first and second track portions; the connector panel includes engagement portions that have drive features coupled to the flocked drive cable within guide rails of the first and second track portions, respectively; the track is coupled to a body defining a storage space; and the support assembly is configured to translate over the storage space between the first and second positions.

(53) According to various examples, a center console includes a body. A track is coupled to the body. A support assembly has side panels rotatably coupled to a connector panel. The side panels are operable between a folded condition over the connector panel and a deployed condition forming a continuous surface with the connector panel. An actuation assembly is operably coupled to the support assembly and the track. The actuation assembly includes a cable extending through the track and coupled to the support assembly. A motor has a drive gear. The drive gear is configured to engage the cable to translate the support assembly between first and second positions relative to the body. Embodiments of the present disclosure may include one or a combination of the following features: the cable is a flocked cable and the drive gear is configured to engage coils of the flocked cable; the body defines a storage space; the support assembly translates over the storage space between the first and second positions; the support assembly is in the folded condition when in the first position on the body and in the deployed condition when in the second position of the body; the side panels are coupled to the connector panel via hinge assemblies, respectively, and wherein the side panels, the connector panel, and the hinge assemblies form a continuous surface when in the deployed condition; and the connector panel includes engagement portions extending proximate to sides of the body within a storage space defined by the body to engage the track.

(54) For purposes of this disclosure, the term “coupled” (in all of its forms, couple, coupling, coupled, etc.) generally means the joining of two components (electrical or mechanical) directly or indirectly to one another. Such joining may be stationary in nature or movable in nature. Such joining may be achieved with the two components (electrical or mechanical) and any additional intermediate members being integrally formed as a single unitary body with one another or with the two components. Such joining may be permanent in nature or may be removable or releasable in nature unless otherwise stated.

(55) It is also important to note that the construction and arrangement of the elements of the invention as shown in the exemplary examples is illustrative only. Although only a few examples of the present innovations have been described in detail in this disclosure, those skilled in the art who review this disclosure will readily appreciate that many modifications are possible (e.g., variations in sizes, dimensions, structures, shapes, and proportions of the various elements, values of parameters, mounting arrangements, use of materials, colors, orientations, etc.) without materially departing from the novel teachings and advantages of the subject matter recited. For example, elements shown as integrally formed may be constructed of multiple parts or elements shown as

multiple parts may be integrally formed, the operation of the interfaces may be reversed or otherwise varied, the length or width of the structures and/or members or connectors or other elements of the system may be varied, the nature or number of adjustment positions provided between the elements may be varied. It should be noted that the elements and/or assemblies of the system might be constructed from any of a wide variety of materials that provide sufficient strength or durability, in any of a wide variety of colors, textures, and combinations. Accordingly, all such modifications are intended to be included within the scope of the present innovations. Other substitutions, modifications, changes, and omissions may be made in the design, operating conditions, and arrangement of the desired and other exemplary examples without departing from the spirit of the present innovations.

(56) Modifications of the disclosure will occur to those skilled in the art and to those who make or use the disclosure. Therefore, it is understood that the embodiments shown in the drawings and described above are merely for illustrative purposes and not intended to limit the scope of the disclosure, which is defined by the following claims, as interpreted according to the principles of patent law, including the doctrine of equivalents.

(57) It is to be understood that variations and modifications can be made on the aforementioned structure without departing from the concepts of the present disclosure, and further it is to be understood that such concepts are intended to be covered by the following claims unless these claims by their language expressly state otherwise.

Claims

1. A vehicle console assembly, comprising: a body defining a storage space, wherein the body includes an upper rim; a track extending along the upper rim of the body; a support assembly operable between a first position at a first end of the upper rim and a second position at a second end of the upper rim, wherein the support assembly is operable between a folded armrest condition and a deployed tray condition, wherein the support assembly includes an engagement portion extending into the storage space, and wherein the engagement portion includes a drive feature slidably received in a guide rail of the track; and an actuation assembly operably coupled to the support assembly, wherein the actuation assembly includes: at least one cable having coils, wherein the at least one cable is coupled to the support assembly and extends through a first channel defined by the track; and a motor configured to engage the coils of the at least one cable to translate the support assembly between the first position and the second position.
2. The vehicle console assembly of claim 1, wherein the support assembly includes: a connector panel extending between opposing sides of the rim; a first side panel rotatably coupled to a first side of the connector panel; and a second side panel rotatably coupled to a second side of the connector panel, wherein the first and second side panels are disposed on the connector panel in the folded armrest condition and disposed adjacent to the connector panel in the deployed tray condition.
3. The vehicle console assembly of claim 1, wherein the support assembly is in the folded armrest condition when in the first position on the upper rim and in the deployed tray condition when in the second position on the upper rim.
4. The vehicle console assembly of claim 1, wherein the at least one cable includes a drive cable and a return cable, and wherein the drive feature is fixedly coupled to the drive cable.
5. The vehicle console assembly of claim 1, further comprising: an electronic control unit communicatively coupled to the actuation assembly, wherein the actuation assembly includes a Hall sensor configured to sense rotations of the motor, and wherein the electronic control unit is configured to monitor sensed rotations to determine a position of the support assembly relative to the upper rim.
6. The vehicle console assembly of claim 4, wherein the drive cable extends through the first

channel defined by the track, and wherein the return cable extends through a second channel defined by the track.

7. The vehicle console assembly of claim 6, wherein the track includes a pair of track portions on opposing sides of the upper rim, and wherein the at least one cable extends through each of the pair of track portions.
8. A power tray system for a center console, comprising: a track including a first track portion and a second track portion spaced from the first track portion; a flocked drive cable extending through the first track portion; a flocked return cable extending through the second track portion, wherein the flocked return cable is connected to the flocked drive cable; a motor having a drive gear configured to engage coils of the flocked drive cable; and a support assembly coupled to the flocked drive cable, wherein the drive gear is configured to engage the coils of the flocked drive cable to drive movement of the flocked drive cable relative to the track and, consequently, translate the support assembly between first and second positions relative to the track, and wherein the support assembly is operable between a folded armrest condition and deployed tray condition.
9. The power tray system of claim 8, wherein the support assembly includes side panels, and wherein each side panel includes a cushion on one side thereof, and wherein the cushions are on upper surfaces of the side panels when the side panels are in the folded armrest condition.
10. The power tray system of claim 9, wherein the cushions are on lower surfaces of the side panels when the side panels are in the deployed tray condition.
11. The power tray system of claim 8, wherein the support assembly includes: a connector panel; and side panels rotatably coupled to the connector panel, wherein the side panels are disposed on the connector panel in the folded armrest condition and adjacent to the connector panel in the deployed tray condition.
12. The power tray system of claim 11, wherein the connector panel extends between the first and second track portions.
13. The power tray system of claim 12, wherein the connector panel includes an engagement portion that has a drive feature coupled to the flocked drive cable within a guide rail of the first track portion.
14. The power tray system of claim 13, wherein the track is coupled to a body defining a storage space, wherein the engagement portion of the connector panel extends into the storage space, and wherein the support assembly is configured to translate over the storage space between the first and second positions.
15. A center console, comprising: a body defining a storage space; a track coupled to the body; a support assembly having side panels rotatably coupled to a connector panel, wherein the side panels are operable between a folded condition over the connector panel and a deployed condition forming a continuous surface with the connector panel, wherein the support assembly includes an engagement portion extending into the storage space, and wherein the engagement portion engages the track; and an actuation assembly operably coupled to the support assembly and the track, wherein the actuation assembly includes: a cable extending through the track and coupled to the support assembly; and a motor having a drive gear, wherein the drive gear is configured to engage the cable to translate the support assembly between first and second positions relative to the body.
16. The center console of claim 15, wherein the cable is a flocked cable and the drive gear is configured to engage coils of the flocked cable.
17. The center console of claim 15, wherein the support assembly translates over the storage space between the first and second positions.
18. The center console of claim 15, wherein the support assembly is in the folded condition when in the first position on the body and in the deployed condition when in the second position on the body.
19. The center console of claim 15, wherein the side panels are coupled to the connector panel via hinge assemblies, respectively, and wherein the side panels, the connector panel, and the hinge

assemblies form a continuous surface when in the deployed condition.

20. The center console of claim 15, wherein the cable extends uninterrupted along a first side of the body, an end of the body, and a second side of the body.
